CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 3: **The World of Metals and Non-metals** | Quick revision, one page

Physical properties of metals

- Lustre: shiny surface. Malleability: can be beaten into sheets. Ductility: can be drawn into wires.
- Sonority: produce a ringing sound. Good conductors of heat and electricity. Generally hard and solid.
- Exceptions: mercury is a liquid metal; sodium and potassium are soft and can be cut with a knife.

Physical properties of non-metals

- Generally dull, brittle, poor conductors of heat and electricity, not sonorous.
- May be solid (sulphur, carbon), liquid (bromine) or gas (oxygen, nitrogen).
- Exceptions: graphite conducts electricity; diamond is the hardest natural substance; iodine is shiny.

Chemical properties

- Metal + oxygen $\rightarrow$ metal oxide (basic: turns red litmus blue), e.g. magnesium oxide.
- Non-metal + oxygen $\rightarrow$ non-metal oxide (acidic: turns blue litmus red), e.g. sulphur dioxide.
- Metal + dilute acid $\rightarrow$ salt + hydrogen gas (burns with a pop sound).
- Displacement: a more reactive metal displaces a less reactive one from its solution (iron nail in copper sulphate).

Rusting and uses

- Rusting of iron needs air (oxygen) and moisture; prevented by painting, oiling, galvanisation (zinc coating).
- Uses: copper in wires, aluminium in foil and utensils, gold and silver in jewellery, iron in construction.

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